



ATLAS ABP SERIES
ASPHALT BATCH
MIX PLANTS

with **Computerised Panel**, bag house
type pollution control unit and Rated
capacity up to **160 TPH @ 3%**
moisture content



SCOPE OF SUPPLY

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General Description

Stationary, Batch type, Fully Automatic Asphalt Mixing Plant – Model ABP 160

Specification Category	Details
Production and Capacity	140 tons per hour (based on 1.6 ton/m ³ density and 150°C output temperature).
	1750 Kg batch with 80 cycles per hour.
	Rated capacity considers 3% moisture content in aggregates.
	Maintain mix temperature with ±5°C precision.
Material Handling and Quality	Aggregates must be clean, free from dirt, and without oversized material to prevent blockages.
	Less than 3% moisture, free from lumps or agglomeration.
	Accurate dosing with preheating for consistency.
	Advanced bag filters or scrubbers to meet environmental norms.
Automation and Control	Fully automatic electro-pneumatic operation with manual override.

	Integrated systems for real-time data acquisition and process control.
	Emergency stop switches and fire suppression systems.
	Energy-efficient burner systems to optimize fuel usage.
Mixing Homogeneity	High-performance twin-shaft or planetary mixers for uniform output.
Maintenance and Scalability	Modular design for quick servicing and centralized lubrication systems.
	Use of wear-resistant materials in critical areas such as mixer linings.
	Provision for capacity upgrades or integration with advanced technologies.
Environmental Compliance	Compliance with Euro, EPA , or equivalent environmental standards.
	Capability to incorporate RAP for sustainable production.
Operational Versatility	Ability to produce mixes of varying grades (e.g., dense graded, open graded).
	Stationary or mobile design options based on project requirements.

Cold Bin Feeder

Component	Specification
Cold Aggregate Hoppers	Four hoppers with a heaped capacity of approx. 40 m ³ (10 m ³ each).
	Wide opening bin width to accommodate various loader capacities.
	Anti-bridging bin design for smooth and consistent material flow, irrespective of material level.
	Sturdy and proven construction for durability.
Belt Feeders	Four feeders, each 500 mm wide and 2.75 m long, with adjustable radial discharge gates.
	Adjustable skirts to prevent leakage between bin bottom and belt surface.
	Feeding capacity: 50 Tons per Hour per hopper.
	Powered by 1.5 kW (2 HP) electric motors for each feeder (total: 4 x 1.5 kW).
Feeding Units	Components include:
	Head and tail pulleys.
	Single roll idlers.
	Endless rubber conveyor.
	Screw-type belt tensioners.
	Replaceable rubber skirting.

	No-flow indicators for monitoring.
	Vibrator on sand bin (0.3 HP or 0.2 kW) for enhanced material flow.
Power and Control	Static inverter variable speed controls with transmission via gear reduction unit.



Parameter	Details
Type	Feeder frame integrated, endless conveyor.
Belt Width	600 mm.
Length	24 meters.
Rated Feeding Capacity	200 Tons per Hour.
Motor Type	5 HP squirrel cage electric motor.
Drive System	Speed reduction unit.
Idlers	Trough idlers, roller guides, and return rollers for smooth belt operation.
Belt Tensioning Mechanism	Screw-type belt tensioners for precise tension adjustment.
Connected Power	5 HP(3.75 Kw)
Construction Material	Heavy-duty construction for durability and reliability.
	Low Height Feeder due to which minimal ramp making therefore easy maintenance and less expenses

Slinger conveyor

Component	Specification
Type	High-speed slinger (throwing) conveyor.
Belt Dimensions	600 mm wide x 18 m long.
Rated Feeding Capacity	200 Tons per Hour (Tph).
Power	5 HP (3.75 kW) squirrel cage electric motor.
Drive Mechanism	Speed reduction unit for controlled and efficient operation.
Features	- Equipped with trough idlers for optimal belt support.
	- Roller guides for proper belt alignment.
	- Return rollers to ensure smooth belt return movement.
	- Mounted on a rigid structural frame for stability and durability.

Dryer Drum

Component	Specification
Type	Cylindrical, inclined, counterflow type dryer drum.
Dimensions	- Drum Diameter: 1,820 mm (1.82 m).
	- Drum Length: 7,000 mm (7.0 m).
	- Drum Slope: 5 degrees for effective material flow.
Temperature	Maximum output temperature: 150°C at the drum outlet.
Drying Capacity	Internal flights are designed to provide higher drying capacity, even with changes in aggregate moisture content.
Heating Efficiency	Internal dryer flights are field-proven and specifically arranged for:
	- Enhanced heating efficiency.

	- Homogeneous heating of aggregates with varying grain sizes.
Support System	- Four heavy-duty trunnion rollers for smooth drum rotation.
	- Two thrust rollers for axial stability.
	- Two trunnion rings with friction wheel drive for consistent operation.
	- Rigid frame construction for durability and stability.
Power	4 x 10 HP (7.5 Kw) motors for friction wheel drive.
Bearings	Spherical roller type for durability and load distribution.
Additional Features	- Hot aggregate temperature measured via a thermocouple-type probe at the dryer outlet.



Burner (Modulating)

Component	Specification
Model	FCMB/LO/M/650
Type	Modulating

Capacity	8.5 Mw
Fuel Type	Diesel / LDO
Burner Rating	350 to 950 liters per hour (lph).
Blower	- Power: 20 HP (15 Kw)
Fuel Pump	- Power: 5 HP (3.75 Kw)
Control	Automatic on/off control for efficient and reliable operation.

Two Stage Dry Dust Collection System (Bag House Filter Unit)

Primary Dust Collecting Unit

Component	Specification
Primary Dust Collection System	Cyclone or pre-separator to trap larger dust particles from the dryer drum exhaust.
	Includes scavenging for the batching tower by screw conveyor.
Type	Screw conveyor for dust transfer.
Conveyor Capacity	5.5 HP (4Kw)
High-Capacity Exhaust Fan	Turbo fan with Variable Frequency Drive (VFD) for enhanced control.
Fan Power	75 HP (55 kW)

Bag House Filter Unit



Component	Specification
Dust Collection System Type	Reverse Air Flow type bag filter.
Number of Bags	252 bags (304 mm x 2100 mm).
Filter Area	330 m ² (approx.).
Bag Material	Meta Aramid felt for high-efficiency filtration.
Normal Temperature Limit	180°C for optimal performance.
Dust Emission	0.1 g/m ³ (Normal operation).
De-dusting System	Fully automatic with 3 de-dusting devices to minimize maintenance of air compressors, solenoid valves, and pneumatic controls.
Safety Features	Safety interlocks (e.g., high exhaust temperature) to ensure safe operation.
Connected Power	1 HP (0.75 kW) per unit, 3 units total.
Capacity	37,000 m ³ /hr air flow capacity.

Fines re-cycling system

Component	Specification
System Type	Fines recycling system with screw conveyor for conveying entrapped fines.
Functionality	The system conveys fines from the bag house to the Hot aggregate elevator, which feeds the filler weigh hopper atop the mixing tower.
Connected Power	10 HP (7.5 kW)



Hot elevator

Specification	Details
Type	Centrifugal discharge type bucket elevator
Capacity	170 TPH (Tons Per Hour)
Connected Power	1 x 15 HP (11 kW)
Motor Power	15 HP
Design	Totally enclosed, vertically mounted
Discharge Mechanism	Centrifugal discharge
Included Components	- Wear plates - Electrical/mechanical drives - Electrical wiring
Safety Feature	Rollback arresting device

Vibrating Screen



Specification	Details
Type	4 decks, inclined circular vibration type
Capacity	160 TPH (Tons Per Hour)
Dimensions (Width x Length)	1500mm wide x 3600mm long
Motor	2 x 6 HP (4.2 kW) vibrating motors
Screen Sieve Sizes	- 3 x 4 mm - 7 x 7 mm - 19 x 19 mm - 25 x 25 mm - 40 x 40 mm
Included Components	- One set of sieves with specified sizes - Duplex Spring absorbers - By-pass system
Design Features	- Inclined circulating, vibrating type - Circular motion to minimize clogging
Sieve Replacement	- Lower two deck sieves hooked at longitudinal ends for easy replacement - No need to remove upper deck sieves for replacement
Maintenance Access	Wide platform around screen for easy and safe exchange or adjustment of sieves and maintenance

Hot aggregates bin



Specification	Details
Number of Compartments	4
Total Storage Capacity	12.0 m ³
Included Features	- Overflow and oversize rejection chute - Pneumatic controlled cut-off gates - Sampling device on each compartment
Level Indicators	One-point bin level indicator in each compartment
Temperature Measurement	Thermocouple-type temperature sensor in bin no. 1

Weighing system

Specification	Details
Weighing System Type	Automatic accumulating and individual weighing type
Aggregate Weigh Hopper	- Accumulating type - Capacity: 2,600 kg (1.6 m³) - 2 discharge gates - Supported by 4 load cells

Filler Weigh Hopper	- Individual weighing type - Capacity: 225 kg (0.28 m³) - Butterfly valve - Supported by 3 load cells
Asphalt Weigh Hopper	- Individual weighing type - Capacity: 180 kg (0.18 m³) - Butterfly valve - Supported by 3 load cells



Pug mill type mixer unit

Specification	Details
Type	Twin shaft jacketed pug mill
Mixing Capacity	1750 kg per batch
Dimensions (L x W x H)	2,230 x 1,170 x 1,270 mm
Connected Power	2 x 25 HP (18.5 kW)
Construction Features	- Welded steel shell- Lined with replaceable wear-resistant spiral liners
Mixing Mechanism	- Specially designed arms and tips for a homogeneous mix in a short time
Discharge Gate	- Wide opening air-operated gate- Operated by 2 cylinders
Drive System	Shaft-mounted gear box
Bearings	Spherical bearings
Truck Clearance	3,480 mm high x 3,700 mm wide



Fines / Filler Feeding System



- 3 Tons capacity filler hopper with screw conveyor power by 7.5 HP (5.5 KW) feeding filler from storage hopper to filler weighing

Asphalt Pumps

A 7.5 kW (10 HP) 450 Lpm Jacketed, gear type asphalt transfer pump is provided for feeding asphalt from the storage tank to the asphalt weigh hopper



Type	Hot oil jacketed gear pump
Pressure	2 kgf/cm ²
Capacity	450 L/min
Motor	10 HP (7.5.KW)

Bitumen Storage And Heating System

Specification	Details
Number of Tanks	2
Capacity (Each)	30,000 liters
Design	Cylindrical, horizontal, fully insulated, and cladded
Features	- Internal heat transfer coils- Temperature indicator



Thermic Oil Heating System	
Type	Normal pressure, oil heating horizontal tubular boiler
Capacity	300,000 kcal/h input
Fuel	High-speed diesel fuel or ASTM D-396 No. 4 fuel oil
Burner	
- Model	FCMP/LO/S/60
- Type	High-pressure mechanical atomizing, gun type oil burner
- Capacity	18 to 60 L/h
- Control	Automatic ignition with high-low control
- Safety Controls	- Flame eye- Low-level cut-out float switch- Pressure switch- High-temperature cut-out thermostat
Piping and Controls	
- Hot Oil Transfer Piping	Total system with necessary control valves
- Bitumen Piping	Jacketed piping with process valves and fittings for manual tank selection and single bitumen delivery pipeline
Hot Oil Transfer Pump	
- Type	Centrifugal pump
- Capacity	30 m ³ /hour
- Head	55 m
- Connected Power	7.5 kW
Auxiliary Power	1 HP

Fuel Tank

18 000 Ltrs capacity, vertical, cylindrical tank for LDO

Control Cabins



Comprising Of Operator's Control Cabin

2500 mm wide x 3000 mm long, Fully Air-conditioned operator's control cabin complete with weather proof exterior and fitted with tinted toughened glass windows and flush entrance door with lock.

Interconnecting

wiring (related to all the above sections) designed to suit the positioning of the unit within 10 m radius of the mixing unit.

Electrical Power Control Console

Complete Motor control console for direct online starting and for star delta starting on necessary devices.

Distribution Switch Board

- Completely enclosed steel cubicle with push buttons starting of motors on star delta starters.
- Magnetic switches with overload thermal relay for each individual motor.
- Safety Interlocks and protection devices.
- Meters for online current voltage measurements for high capacity motors.
- Emergency stop button.
- Short circuit protection with switch fuse unit.
- Inverter Drives for control of cold aggregates bin feeders

Temperature Indication

- 0 -300 Deg C, Electronic multi point scanner for measuring:
- Aggregates temperature measuring in Hot Bin no. 1.
- Asphalt feeding temperature.
- Bag filter inlet gas temperature.
- Bitumen tank temperature. (tank # - 1 & 2)

Sequence Control panel

- Programmable Logic Control (PLC) with man machine interface.
- Fully automatic electronic solid state sequence control.
- Operator override for manual or semi automatic control.
- Timer controls for both wet and dry cycle.
- Pilot lamps for indication of sequence currently operated.
- Signals and alarms.
- Storage of Fifty different type of mix proportions.
- Automatic free-fall compensation.
- Direct dialing of mix design by percentage.
- Preset batch counter controls the number of batches for a truck-full load.
- Provision for printing operating data on a day to day basis, like - Mix proportions, Batch weight, Total no. of batches, Subtotal, gross total etc.

Dryer & Burner control.

- Hot aggregates temperature at dryer discharge end.
- Remote Burner flame synchronization with hot aggregates temperature.
- Safety interlock, in case of high temperature.

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OPTIONAL ACCESSORIES TO MAKE YOUR PLANT AFFORDABLE

Accessory	Functionality	Technical Details
Cold RAP (up to 40%)	Allows the use of recycled asphalt materials in the mix without preheating.	- RAP typically contains 3-6% residual bitumen and aggregates. - Efficient blending ensures mechanical and thermal stability.
Hot RAP – HRC (up to 60%)	Incorporates pre-heated recycled asphalt materials into the mix.	- Requires precise temperature control to prevent bitumen oxidation. - Can maintain the binder's rheological properties.
Air Atomized Burner	Utilizes compressed air to atomize fuel for the drying drum.	- Atomization improves fuel droplet surface area for combustion. - Operates with high thermal efficiency (~90%).
Coal Burner	Uses coal as a fuel source for the drying drum.	- Requires pulverized coal for uniform combustion. - Emission control measures like scrubbers or ESPs are recommended.
Gas Burner (for Bitumen Tank)	Uses LPG or natural gas to heat the bitumen tank.	- Gas combustion achieves >95% efficiency. - Reduced NOx and SOx emissions compared to liquid fuels.
CNG Burner (for Thermic Oil Heater)	Employs compressed natural gas to heat thermic oil used for asphalt plant operations.	- Methane (CH₄) in CNG burns at a high calorific value (~50 MJ/kg). - Produces 30-40% fewer emissions than coal or oil burners.

CNG Burner (for Drum Burner of Batch Asphalt Plant)	Heats the aggregate in batch asphalt plants using CNG.	- Requires proper burner calibration for optimal mixing of gas and air. - Generates consistent flame temperature (~1,900°C).
Thermic Oil Heater with Coal/Wood Fire Burner	Provides heating using thermic oil heated by burning coal or wood.	- Efficient thermic fluid systems achieve 85-90% heat transfer efficiency. - Requires a well-insulated heating coil for longevity.

OTHER ANCILLIARY ITEM

Ancillary Item	Functionality	Technical Details
Foam Bitumen	Creates foamed bitumen by injecting water into hot bitumen, expanding its volume and improving mix workability.	- Bitumen expands up to 10-15 times its volume. - Optimal water content (~2%) and temperature (~160-180°C) critical for foam stability.
Stone Mastic Asphalt (SMA)	A high-performance asphalt mix with a stone skeleton and bitumen-mastic matrix for heavy-duty surfaces.	- SMA typically includes stabilizing additives like cellulose or polymer fibers to prevent bitumen drain-off.
Liquid Additive	Enhances bitumen properties for better adhesion, flexibility, and temperature resistance.	- Additives like amines or polymers chemically modify bitumen to enhance adhesion and aging resistance.
Inline Silo (25-300 Tons, 1-3 Compartments)	Storage solution for aggregates or finished products, ensuring segregation and availability.	- Designed for easy loading and retrieval, with compartments preventing material mixing. - Load cells ensure accurate material tracking.
Hoist for Easy Maintenance (Elevator & Screen)	Mechanical lifting system for accessing and maintaining critical plant components.	- Typically uses electric or hydraulic systems with lifting capacities suited for heavy plant components.
Dust Conditioner	Mixes water with collected dust to form a slurry for safe disposal or recycling.	- Uses water spray systems for consistent wetting. - Recycled water systems enhance sustainability.

